

FIRST YEAR ENGG - F.E.

Programming and Problem Solving

(2024 Credit Pattern) (Semester - I/II)

UNIT I: Problem Solving, Programming and Python Programming

1. HIGHEST IMPORTANCE TOPICS:

a) Problem Solving Concepts

- Types of problems (Q1-a: 3 marks)
- Problem solving definition & steps (Q1-d: 4 marks)
- Top-down design (Q2-d: 4 marks)

b) Python Programming Basics

- Features of Python (Q2-c: 4 marks)
- History of Python & software developed (Q1-b: 3 marks)
- Applications of Python (Q2-b: 3 marks)

c) Object Oriented Programming

- Features of OOP (Q1-c: 4 marks)

2. MEDIUM IMPORTANCE TOPICS:

a) Flowcharts

- Flowchart symbols and usage (Q2-a: 3 marks)

b) Programming Paradigm

- General programming paradigms

3. LEAST/NOT APPEARED TOPICS:

- Problem solving in everyday life (detailed theory)
- Difficulties with problem solving
- Problem solving aspects (detailed)

- Future of Python (specific predictions)

UNIT II: Advance Data Types and Decision Control Statements

1. HIGHEST IMPORTANCE TOPICS:

a) Advanced Data Types

- Tuples, Lists, Sets, Dictionary - List all & explain any two (**Q3-a: 3 marks**)
- Dictionary operations (**Q4-d: 4 marks** - What is dictionary + 3 operations)

b) Decision Control Statements

- if, if-else, nested if, if-elif-else (**Q3-d: 4 marks**)
- Programs: positive/negative/zero check (**Q3-c: 4 marks**)
- Programs: even/odd check (**Q4-c: 4 marks**)

c) Loop Control Terms

- break, continue, pass, range (**Q3-b: 3 marks** - explain any 3)

d) Operators

- Arithmetic operators (**Q4-a: 3 marks**)

2. MEDIUM IMPORTANCE TOPICS:

a) Loops Comparison

- for loop vs while loop (**Q4-b: 3 marks**)

b) Loop Structures

- while loop, for loop basics
- Selecting appropriate loop

3. LEAST/NOT APPEARED TOPICS:

- Nested loops (detailed implementation)
- else statement used with loops (specific questions)
- Detailed comparison of all data types

UNIT III: Functions and Strings

1. HIGHEST IMPORTANCE TOPICS:

a) Functions

- Regular vs Lambda functions (**Q5-a: 4 marks**)
- return statement usage (**Q5-b: 5 marks**)
- Variable scope (local & global) and lifetime (**Q6-b: 5 marks**)

b) String Operations

- ord() and chr() functions (**Q6-a: 4 marks**)
- String concatenation program (**Q5-c: 5 marks** - algorithm & program)
- String reversal program (**Q6-c: 5 marks** - using user-defined function)

2. MEDIUM IMPORTANCE TOPICS:

a) Function Basics

- Need for functions
- Function definition and call
- Documentation string
- Good programming practices

b) String Operations

- String slicing
- String formatting operator
- Built-in string methods

3. LEAST/NOT APPEARED TOPICS:

- Introduction to modules (detailed)
- Introduction to packages in Python
- Standard library modules
- Strings are immutable (specific question)
- String multiplication and appending

- in and not in operators for strings
- Comparing strings (detailed)
- Iterating strings
- The string module

UNIT IV: File Handling and Dictionaries

1. HIGHEST IMPORTANCE TOPICS:

a) File Handling

- Purpose & use of file handling (**Q7-a: 4 marks**)
- Types of files (**Q8-a: 4 marks**)
- Program to append data to existing file (**Q7-c: 5 marks**)
- Program for absolute path using os.path.join (**Q8-c: 5 marks**)

b) Directory Methods

- mkdir(), rmdir(), listdir(), chdir() (**Q8-b: 5 marks** - syntax & usage)

c) Dictionaries

- Mutable vs Immutable data types focusing on dictionaries (**Q7-b: 5 marks**)

2. MEDIUM IMPORTANCE TOPICS:

a) File Operations

- File path
- Opening and closing files
- Reading and writing files
- File positions
- Renaming and deleting files

b) Dictionary Operations

- Creating, accessing, adding, updating values

3. LEAST/NOT APPEARED TOPICS:

- Case Study (explicitly excluded from theory exam as per syllabus)
- Detailed dictionary methods (beyond basic operations)

UNIT V: Object Oriented Programming

1. HIGHEST IMPORTANCE TOPICS:

a) Classes and Objects

- Classes and objects detailed explanation (**Q9-c: 5 marks**)
- Class variables vs Object variables (**Q9-a: 4 marks**)

b) Special Methods

- **init()** and **del()** methods in object lifecycle (**Q10-a: 4 marks**)

c) Access Modifiers

- Public vs Private members (**Q10-b: 5 marks** - with examples)

d) OOP Principles

- Data abstraction & Encapsulation (**Q10-c: 5 marks**)

e) Memory Management

- Garbage collection in Python (**Q9-b: 5 marks** - significance & how Python handles it)

2. MEDIUM IMPORTANCE TOPICS:

a) OOP Features

- Inheritance, polymorphism, containership
- Reusability, delegation

b) Class Components

- class method and self-argument
- Built-in functions (Get, Set, Delete attributes)

- Class methods
- Static methods

3. LEAST/NOT APPEARED TOPICS:

- Structured vs Object Oriented (detailed comparison)
- Methods and message passing (specific questions)
- Detailed containership concept
- Delegation (detailed theory)

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MOST EXPECTED QUESTIONS unit-wise based on PYQ analysis and syllabus weightage:

UNIT I: Problem Solving, Programming and Python Programming

3 Marks Questions:

1. What are the types of problems? Explain in detail.
2. Explain brief history of Python programming language.
3. Write down the applications of Python programming languages.
4. List the features of Python programming language.
5. Describe flowchart symbols used in problem solving.

4 Marks Questions:

1. Define problem solving. Write down steps of problem solving process.
2. Describe object oriented programming features in detail.
3. Explain the Python programming features with examples.
4. Describe TOP-DOWN design approach in detail with example.
5. Differentiate between procedural and object-oriented programming paradigm.

UNIT II: Advance Data Types and Decision Control Statements

3 Marks Questions:

1. List down all the advanced data types in Python & Explain any two.
2. Describe the following terms with examples (any three): break, continue, pass, range
3. Explain different arithmetic operators with suitable examples.
4. Write down the difference between for loop and while loop.
5. Differentiate between tuple and list with examples.

4 Marks Questions:

1. Explain decision control statements in Python in detail (if, if-else, nested if, if-elif-else).

2. Write a Python program to check whether a number entered by user is positive, negative, or zero.
3. Develop a Python program to check whether a given number is even or odd.
4. What is the dictionary data type? Explain any 3 operations of the dictionary data type.
5. Write a program to find the largest of three numbers using nested if statements.
6. Explain while loop and for loop with suitable examples.

UNIT III: Functions and Strings

4 Marks Questions:

1. Compare Regular functions and Lambda functions in Python with suitable examples.
2. Analyze the difference between ord() and chr() functions with examples.
3. Explain the need for functions in Python with advantages.
4. What is a module in Python? How to import and use modules?

5 Marks Questions:

1. Describe the use of the return statement in functions with examples.
2. Explain the concepts of variable scope (local and global) and lifetime in Python with suitable code examples.
3. Develop an algorithm & program to concatenate two strings using + operator.
4. Implement a program to reverse a string using user-defined function.
5. Write a program to count vowels and consonants in a string.
6. Explain string slicing operations with examples and write a program to demonstrate it.
7. What are built-in string methods? Explain any 5 with examples.

UNIT IV: File Handling and Dictionaries

4 Marks Questions:

1. Describe the purpose & use of file handling in Python.
2. Explain different types of files in detail (text files and binary files).
3. What is file path? Explain absolute and relative path with examples.
4. Differentiate between reading and writing modes in file handling.

5 Marks Questions:

1. Differentiate between mutable and immutable data types with focus on dictionaries in Python. Provide examples.
2. Explain key directory methods: mkdir(), rmdir(), listdir(), and chdir() with syntax and usage examples.
3. Implement a program to append data to an already existing file.
4. Develop a program to print the absolute path of a file using os.path.join.
5. Write a program to read content from one file and write it to another file.
6. Explain dictionary operations: creating, accessing, adding, updating, and deleting with program examples.
7. Write a program to count the number of words and lines in a text file.

UNIT V: Object Oriented Programming

4 Marks Questions:

1. Differentiate between class variables and object variables with an example.
2. Explain the role of init() and del() methods in object lifecycle.
3. What is a class? Explain class and object with suitable example.
4. Differentiate between class method and static method with examples.

5 Marks Questions:

1. Analyze the significance of garbage collection in Python. How does Python handle object destruction?
2. Describe classes and objects in detail with program example.
3. Distinguish between public and private members of a class with proper examples.
4. Describe data abstraction & Encapsulation in detail with examples.
5. Explain inheritance in Python with types and program examples.

6. **What is polymorphism? Explain method overloading and method overriding with examples.**
7. **Write a program to demonstrate the use of init() method and create multiple objects.**

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